

The image displays a sequence of Feynman diagrams representing a mathematical identity. It begins with the sum of four terms, each featuring a vertical red line and two horizontal blue lines. The first term shows a fermion loop with external momenta p_1 and p_2 , and a wavy line with momentum k . The subsequent three terms show variations in the internal fermion and wavy line momenta, including $k-p_1$, $k-p_2$, and k . The sum of these four terms is equated to the product of two disc integrals, $\text{disc}_{p_1^2} \text{disc}_{p_2^2}$. To the right, a diagram of a disc with two internal vertices and a red line connecting them is shown, with momenta p_1 and p_2 indicated.

$$\begin{aligned}
 & \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} = \text{disc}_{p_1^2} \text{disc}_{p_2^2} \\
 & \text{Disc Diagram}
 \end{aligned}$$